



HOW TO USE THIS SET

20 questions, mixed difficulty, 5 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.

EASY

Which segment of the small intestine has leaf-shaped villi and Brunner's glands in the submucosa?

- A. Jejunum
- B. Sigmoid colon
- C. Duodenum
- D. Cecum
- E. Ileum

2.

EASY

Which crypt cell releases lysozyme, phospholipase A2 and defensins from its granules?

- A. Enteroendocrine cell
- B. Goblet cell
- C. M cell
- D. Paneth cell
- E. Absorptive cell

3.

EASY

Which segment shows the tallest and most numerous plicae circulares, along with long finger-shaped villi?

- A. Duodenum
- B. Rectum
- C. Ileum
- D. Cecum
- E. Jejunum

4.

EASY

Peyer's patches (lymphoid nodules in the lamina propria) are characteristic of which segment?

- A. Ileum
- B. Cardia
- C. Duodenum
- D. Esophagus
- E. Jejunum

5.

MEDIUM

CASE

A biopsy shows leaf-shaped villi, relatively few goblet cells, and glands in the submucosa secreting alkaline mucus.

This segment is the:

- A. Duodenum
- B. Ileum
- C. Jejunum
- D. Cecum
- E. Colon

6.

MEDIUM

CASE

A biopsy shows club-shaped villi, abundant goblet cells, and a lymphoid nodule bulging from the submucosa into the mucosa.

This segment localises to the:

- A. Stomach
- B. Colon
- C. Duodenum
- D. Ileum
- E. Jejunum

7.

MEDIUM

A specialised epithelial cell overlying a Peyer's patch, with a discontinuous basement membrane, that endocytoses antigens and transports them to underlying macrophages is the:

- A. Stem cell
- B. Paneth cell
- C. Goblet cell
- D. Absorptive cell
- E. M cell

8.

MEDIUM

Absorptive (columnar) cells of the small intestine are characterised by:

- A. Mucin-distended cytoplasm with a goblet shape
- B. Location exclusively at the crypt base
- C. A discontinuous basement membrane
- D. A brush border of approximately 3,000 microvilli per cell, maximising surface area for absorption
- E. Basophilic cytoplasm with eosinophilic apical granules

9.

MEDIUM

Which structure permanently increases mucosal + submucosal surface area, is most prominent in the jejunum, and slows chyme transit to allow more absorption time?

- A. Peyer's patches
- B. Crypts of Lieberkühn
- C. Microvilli
- D. Villi
- E. Plicae circulares

10.

MEDIUM

Where are stem cells located within the intestinal crypt, and what is their function?

- A. At the villus tip; absorption
- B. At the crypt opening; mucus secretion
- C. Within the muscularis mucosae; contraction
- D. In the lamina propria; immune surveillance
- E. At the base of the crypt; undifferentiated, highly mitotic precursors of all other crypt/villus cell types

11.

MEDIUM

Which cell type increases in density progressively toward the ileum, interspersed among absorptive cells?

- A. Goblet cells
- B. Enteroendocrine cells
- C. Stem cells
- D. M cells
- E. Paneth cells

12.

MEDIUM

Enteroendocrine cells of the small intestinal crypt secrete which of the following?

- A. Digestive enzymes for luminal digestion
- B. Mucus for lubrication
- C. Lysozyme and defensins
- D. Antibodies for mucosal immunity
- E. Serotonin, somatostatin, CCK, and secretin, regulating motility, secretion and appetite

13.

MEDIUM

Which wall layer of the small intestine contains Brunner's glands, and in which segment are they found?

- A. Muscularis externa, in the jejunum
- B. Lamina propria, in the ileum
- C. Mucosa, in the ileum
- D. Submucosa, in the duodenum only
- E. Serosa, in all segments

14.

MEDIUM

What is the approximate turnover time of the small intestinal epithelium, driven by crypt stem cell proliferation?

- A. 10-14 days
- B. Several months
- C. Turnover does not occur in adult epithelium
- D. 2-4 days
- E. 6-12 hours

15.

HARD

CASE

An unlabelled biopsy shows long, finger-shaped villi with an intermediate density of goblet cells, tall plicae circulares, but no Brunner's glands and no Peyer's patches visible in this particular section.

Which segment is most consistent with this combination of findings?

- A. Esophagus, since villi may be mistaken for esophageal papillae
- B. Cecum, since it can show villi in disease states
- C. Ileum, since Peyer's patches are not always visible on a given section
- D. Jejunum, since long finger-shaped villi with prominent plicae circulares and intermediate goblet cell density, without Brunner's glands or Peyer's patches, is the jejunal pattern
- E. Duodenum, since Brunner's glands may simply be absent from this particular field

16.

HARD

A drug selectively destroys Paneth cells while sparing all other crypt and villus cell types. Which specific protective function would be lost, and which cell populations would remain intact?

- A. No functional consequence, since Paneth cells are vestigial in humans
- B. Loss of absorption, with goblet and stem cells intact
- C. Loss of M cell antigen transport, since Paneth and M cells are the same lineage
- D. Loss of antimicrobial granule release (lysozyme, phospholipase A2, defensins) that helps regulate the crypt microenvironment, while absorptive, goblet, stem, and enteroendocrine cells remain structurally intact
- E. Loss of mucus secretion entirely, since Paneth cells are the sole mucus source

17.

HARD

CASE

A researcher compares antigen sampling capacity between a segment rich in Peyer's patches and a segment without them.

Which cell type would be expected to be present in the Peyer's patch-rich segment but essentially absent elsewhere, and what distinguishes its ultrastructure?

- A. Paneth cells, distinguished by eosinophilic granules
- B. Absorptive cells, distinguished by a brush border
- C. M cells, distinguished by a discontinuous basement membrane that facilitates antigen transport to underlying lymphoid tissue
- D. Stem cells, distinguished by high mitotic activity
- E. Goblet cells, distinguished by mucin vacuoles

18.

HARD

If a biopsy from the duodenum shows normal villus architecture but complete absence of Brunner's glands in the submucosa, which functional consequence would be most directly expected?

- A. Reduced buffering of acidic chyme entering the duodenum, since Brunner's glands normally secrete alkaline mucus to protect the mucosa and optimise pancreatic enzyme activity
- B. Complete loss of goblet cell mucus secretion
- C. Loss of lymphoid surveillance in the duodenum
- D. Failure of epithelial turnover
- E. Loss of nutrient absorption entirely

19.

HARD

CASE

A 24-year-old with longstanding untreated celiac disease undergoes duodenal biopsy showing blunted (flattened) villi with crypt hyperplasia.

Which structural consequence follows most directly from villus blunting, in terms of the surface-area principles described in this lecture?

- A. Increased plicae circulares formation to compensate
- B. Increased surface area for absorption, compensating for the disease
- C. Reduced absorptive surface area, since villi are one of the four surface-area-maximising adaptations (length, plicae circulares, villi, microvilli) of the small intestine
- D. No change in absorptive capacity, since crypts compensate fully for villus loss
- E. Complete loss of the muscularis mucosae

20.

HARD

Which combination of histologic features would most reliably differentiate a duodenal biopsy from a jejunal biopsy, integrating villus shape, glandular structures, and serosal covering?

- A. The presence of a muscularis externa is unique to the jejunum
- B. Villus shape alone is sufficient and no other feature is needed
- C. Peyer's patches distinguish duodenum from jejunum
- D. Goblet cell density alone reliably separates duodenum from jejunum
- E. Leaf-shaped villi plus Brunner's glands in the submucosa plus a serosa-and-adventitia covering together point to duodenum, versus jejunum's long finger-shaped villi, tallest plicae circulares, and pure serosal covering without Brunner's glands

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	C	Duodenum
2	D	Paneth cell
3	E	Jejunum
4	A	Ileum
5	A	Duodenum
6	D	Ileum
7	E	M cell
8	D	A brush border of approximately 3,000 microvilli per cell, maximising surface area for absorption
9	E	Plicae circulares
10	E	At the base of the crypt; undifferentiated, highly mitotic precursors of all other crypt/villus cell types
11	A	Goblet cells
12	E	Serotonin, somatostatin, CCK, and secretin, regulating motility, secretion and appetite
13	D	Submucosa, in the duodenum only
14	D	2-4 days
15	D	Jejunum, since long finger-shaped villi with prominent plicae circulares and intermediate goblet cell density, without Brunner's glands or Peyer's patches, is the jejunal pattern
16	D	Loss of antimicrobial granule release (lysozyme, phospholipase A2, defensins) that helps regulate the crypt microenvironment, while absorptive, goblet, stem, and enteroendocrine cells remain structurally intact
17	C	M cells, distinguished by a discontinuous basement membrane that facilitates antigen transport to underlying lymphoid tissue
18	A	Reduced buffering of acidic chyme entering the duodenum, since Brunner's glands normally secrete alkaline mucus to protect the mucosa and optimise pancreatic enzyme activity
19	C	Reduced absorptive surface area, since villi are one of the four surface-area-maximising adaptations (length, plicae circulares, villi, microvilli) of the small intestine
20	E	Leaf-shaped villi plus Brunner's glands in the submucosa plus a serosa-and-adventitia covering together point to duodenum, versus jejunum's long finger-shaped villi, tallest plicae circulares, and pure serosal covering without Brunner's glands